DOOR LOCK PROJECT

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introduction

The primary function of this project is to create a secure door lock system that can be controlled through a keypad interface. The AT89S52 microcontroller serves as the central processing unit, managing input from the keypad for unlocking the door and controlling the motor to physically lock or unlock the door. A buzzer provides audible feedback during operation, and an LCD displays system status and user prompts.

Components and Interfaces:

1. AT89S52 Microcontroller:

- Description: The AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller with 8K bytes of in-system programmable Flash memory. It is a core component for handling the logic and control functions of the system.
- Interface: Interacts with the keypad for input, drives the LCD for output, controls the motor through the H-bridge, and activates the buzzer based on system states.

2. LCD Display:

- Description: Typically, a 16x2 LCD is used to display information such as error messages and prompts for the user.
- Interface: Connected to the microcontroller via a parallel interface. It receives commands and data from the microcontroller to update the display.

3. 4x4 Keypad:

- Description: A matrix keypad with 16 keys that allows users to input their access code and the buttons can do many functions.
- Interface: Wired to the microcontroller using row and column connections. The microcontroller scans the keypad by activating one row at a time and reading the columns to detect pressed keys.

4. Motor:

- Description: A DC motor used to actuate the lock mechanism.
- Interface: Controlled by the microcontroller via an H-bridge, which allows for direction control (locking or unlocking) and power handling.

5. H-Bridge (Motor Driver):

- Description: An electronic circuit that enables a voltage to be applied across a load in either direction. Essential for driving the DC motor.
- Interface: Receives control signals from the microcontroller to change the polarity of the voltage applied to the motor, thus controlling the motor's direction.

6. Buzzer:

- Description: Provides auditory feedback when an incorrect passcode is entered.
- Interface: Driven by a digital output from the microcontroller, often through the H-bridge switch to handle the current requirements.

7. Toggle Switch:

- Description: Activate an interruption to change the password.

Design:

